

Model-Aware Positive Stieltjes Quadrature: From Structural Minimax to Gauss–Legendre

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Abstract

Generalized Gaussian and rationally exact quadrature already exploit non-polynomial structure, while rational minimax approximation under interpolation constraints is an established approximation problem. We consider the more specific design of positive symmetric Stieltjes-response quadrature rules that minimize relative error uniformly over a continuum of Lorentzian widths while matching a variable prefix of polynomial moments. Within this class, n positive node pairs and moment depth k give the feasible range $0 \leq k \leq 2n$, an admissible dimension $2n - k$, and the Gauss–Legendre rule at $k = 2n$. For $k < 2n$, a difference-numerator degree bound yields a $2n - k + 1$ -point alternation certificate. High-precision hierarchies for $n = 2, 3, 4$ quantify the trade-off between structural accuracy and polynomial protection. For a smooth residual with Chebyshev coefficients $|a_j| \leq C\rho^{-j}$, an exact coefficient-space norm combines with the Lorentzian error to give the model-aware risk $G_k(\rho) + \tau S_k$. Conservative numerical enclosures assign a unique rule on 97.9067% of the tested parameter domain. Seventy-two multi-start reconstructions recover the upper-half rules, and a shifted-Lorentzian stress test supports local rather than universal robustness. The distinguishing contribution is the combined positive minimax hierarchy, exact selector, and screened decision map, not positivity, moment matching, or rational approximation separately.

Keywords: numerical quadrature, Stieltjes functions, rational minimax approximation, moment constraints, Gauss–Legendre quadrature, numerical verification

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1. Introduction

Polynomial exactness is a central organizing principle of numerical quadrature, but it is not the only way to spend a fixed node budget. When the dominant difficulty is a nearby pole, quadrature may instead incorporate rational functions chosen to match that structure [1]. Generalized Gaussian quadrature extends positive exactness to Chebyshev systems and is naturally described through moment-space geometry [2]. Positive interpolatory formulas and optimization-based moment-matching constructions provide additional general frameworks [6, 7, 8].

A second neighboring direction begins with rational approximation. Modern algorithms compute rational minimax approximants through adaptive barycentric representations and Remez-type iterations [3]. Interpolation-constrained rational minimax approximation, including higher-order constraints, is also established [4]. For Markov and Stieltjes functions, relative rational approximation has a substantial theory [9]. Most directly, quadrature formulas can be derived from rational approximations of Cauchy transforms, with poles and residues becoming nodes and weights and Gauss quadrature connected to Padé approximation at infinity [5].

These results rule out several broad novelty claims: pole-aware quadrature, positive exact quadrature, rational minimax, equioscillation, and constrained rational approximation are not new separately. The narrower problem studied here is to combine them in a model-aware design. At fixed node-pair count, we require a positive symmetric Stieltjes response, match a variable prefix of polynomial moments, and minimize relative error uniformly over a continuum of Lorentzian widths. The resulting family spans a structural minimax rule and the Gauss–Legendre endpoint. The practical question is then not only how to compute the rules, but how to select the moment depth when a smooth uncertain residual is superposed on the structural component.

Within this stated class, the paper makes the following contributions:

- (i) it formulates the positive variable-moment continuum-minimax problem and characterizes its feasible range, dimension, and Gauss–Legendre endpoint;
- (ii) it gives a short class-specific degree/zero-count argument leading to a $2n - k + 1$ -point alternation certificate for every nonzero-dimensional level;

- (iii) it computes the complete path for $n = 2, 3, 4$, including the upper-half levels between the earlier lower-half designs and the Gaussian endpoint;
- (iv) it derives an exact mixed structural/smooth risk and uses it to select the moment depth across a two-parameter model domain;
- (v) it supplies a conservative screened decision map, a shifted-pole scope test, and multi-start reconstruction from perturbed initializations.

The feasible-range theorem and the Gaussian endpoint are important structural components of the framework, but are not claimed as stand-alone inventions of moment or generalized Gaussian theory. A targeted literature search did not identify a direct source combining positivity, continuum Lorentzian minimax, variable moment depth, an exact mixed selector, and a screened parameter partition. This is not an exhaustive proof of novelty; specialist prior-art review remains necessary. The claims are restricted to the symmetric positive class and do not imply universal superiority over Gaussian, adaptive, or problem-specific quadrature.

2. Problem formulation

Let

$$I[f] = \int_{-1}^1 f(x) dx, \quad L_b(x) = \frac{1}{x^2 + b^2}, \quad b \in B = [0.08, 0.12].$$

Set

$$S = \{b^2 : b \in B\} = [0.08^2, 0.12^2].$$

With $s = b^2$ and $y = x^2$,

$$F(s) = I[L_{\sqrt{s}}] = \int_0^1 \frac{y^{-1/2}}{s + y} dy = \frac{2}{\sqrt{s}} \arctan \frac{1}{\sqrt{s}}, \quad s \in S.$$

For a fixed integer $n \geq 1$, define the ordered positive parameter domain

$$\mathcal{P}_n = \{(y, w) \in \mathbb{R}^{2n} : 0 < y_1 < \cdots < y_n < 1, \quad w_j > 0\}.$$

Each $(y, w) \in \mathcal{P}_n$ determines the symmetric rule

$$Q_{y,w}[f] = \sum_{j=1}^n w_j \{f(-\sqrt{y_j}) + f(\sqrt{y_j})\}.$$

Its response on the structural family is

$$r_{y,w}(s) = Q_{y,w}[L_{\sqrt{s}}] = 2 \sum_{j=1}^n \frac{w_j}{s + y_j}.$$

For an integer $k \geq 0$, let

$$m^{(k)} = \left(\frac{1}{2\ell + 1} \right)_{\ell=0}^{k-1}$$

and define the moment map

$$M_k(y, w) = \left(\sum_{j=1}^n w_j y_j^\ell \right)_{\ell=0}^{k-1}.$$

For $k = 0$, both vectors are empty. The admissible parameter set and the corresponding rule class are

$$\Theta_{n,k} = \{(y, w) \in \mathcal{P}_n : M_k(y, w) = m^{(k)}\}, \quad \mathcal{A}_{n,k} = \{Q_{y,w} : (y, w) \in \Theta_{n,k}\}.$$

Lemma 1 (Equivalent interpretations of the moment constraints). *Let $k \geq 1$ and $(y, w) \in \mathcal{P}_n$. The following statements are equivalent:*

- (a) $(y, w) \in \Theta_{n,k}$;
- (b) $Q_{y,w}$ is exact for every polynomial of degree at most $2k - 1$;
- (c)

$$r_{y,w}(s) - F(s) = O(s^{-k-1}) \quad \text{as } s \rightarrow +\infty.$$

Proof. Symmetry gives exactness for odd polynomials. For even monomials,

$$Q_{y,w}[x^{2\ell}] = 2 \sum_{j=1}^n w_j y_j^\ell, \quad I[x^{2\ell}] = \frac{2}{2\ell + 1}.$$

This proves the equivalence of (a) and (b).

For every fixed $m \geq 1$, uniformly for $y \in [0, 1]$,

$$\frac{1}{s + y} = \sum_{\ell=0}^{m-1} (-1)^\ell y^\ell s^{-\ell-1} + O(s^{-m-1}).$$

Consequently,

$$F(s) = 2 \sum_{\ell=0}^{m-1} \frac{(-1)^\ell}{2\ell+1} s^{-\ell-1} + O(s^{-m-1}),$$

whereas

$$r_{y,w}(s) = 2 \sum_{\ell=0}^{m-1} (-1)^\ell \left(\sum_{j=1}^n w_j y_j^\ell \right) s^{-\ell-1} + O(s^{-m-1}).$$

Taking $m = k$ proves that (a) implies (c). Conversely, if (c) holds, uniqueness of the coefficients in these finite asymptotic expansions forces equality of the first k moments, proving (c) implies (a). $\square$

Lemma 2 (Uniqueness of the representation). *If $(y, w), (\hat{y}, \hat{w}) \in \mathcal{P}_n$ and*

$$r_{y,w}(s) = r_{\hat{y},\hat{w}}(s)$$

on a nonempty open real interval, then

$$(y, w) = (\hat{y}, \hat{w}).$$

Proof. The identity theorem makes the two rational functions identical. Their simple poles are $-y_j$ and $-\hat{y}_j$, and the corresponding residues are $2w_j$ and $2\hat{w}_j$. Equality of poles and residues, together with the imposed ordering, gives equality of the parameter vectors. $\square$

3. Feasible-range characterization

Theorem 3 (Feasible range, regular dimension, and Gaussian endpoint). *For $n \geq 1$, the following statements hold:*

- (i) $\Theta_{n,k} \neq \emptyset$ for every integer $0 \leq k \leq 2n$;
- (ii) for $0 \leq k \leq 2n$, $\Theta_{n,k}$ is an embedded smooth submanifold of the open set $\mathcal{P}_n$, of dimension
$$\dim \Theta_{n,k} = 2n - k;$$
- (iii) $\Theta_{n,k} = \emptyset$ for every integer $k \geq 2n + 1$;
- (iv) $\Theta_{n,2n}$ is a singleton, and its associated symmetric rule is the classical $2n$ -point Gauss–Legendre rule.

Proof. Let $d\mu(y) = y^{-1/2} dy$ on $[0, 1]$. By the standard Gaussian quadrature theorem for a positive measure with infinite support, there exists a unique n -node rule with positive weights that is exact for all polynomials of degree at most $2n - 1$. Write its nodes and weights as y_j and $2w_j$. Then

$$\sum_{j=1}^n w_j y_j^\ell = \frac{1}{2} \int_0^1 y^\ell d\mu(y) = \frac{1}{2\ell + 1}, \quad 0 \leq \ell \leq 2n - 1.$$

Thus $\Theta_{n,2n} \neq \emptyset$, and every lower moment prefix is also feasible. The case $k = 0$ is simply $\Theta_{n,0} = \mathcal{P}_n$.

For $1 \leq k \leq 2n$, suppose a linear combination of the rows of $DM_k(y, w)$ vanishes, and define

$$p(t) = \sum_{\ell=0}^{k-1} c_\ell t^\ell.$$

The columns corresponding to w_j give $p(y_j) = 0$. The columns corresponding to y_j give $w_j p'(y_j) = 0$, and positivity gives $p'(y_j) = 0$. Hence every y_j is a double root of p . Since $\deg p \leq k - 1 \leq 2n - 1$, this is possible only when $p \equiv 0$. Therefore DM_k has row rank k at every point of $\mathcal{P}_n$. The regular-level-set theorem gives the stated embedded-manifold dimension.

If $k \geq 2n + 1$, the constraints include exactness for the transformed polynomial

$$g(y) = \prod_{j=1}^n (y - y_j)^2,$$

which has degree $2n$, is nonnegative, and is not identically zero. The discrete rule gives

$$2 \sum_{j=1}^n w_j g(y_j) = 0,$$

whereas

$$\int_0^1 g(y) d\mu(y) > 0,$$

a contradiction.

It remains to identify the endpoint uniquely. Let

$$\pi_n(y) = \prod_{j=1}^n (y - y_j)$$

for any element of $\Theta_{n,2n}$. For every polynomial q of degree at most $n - 1$, exactness through degree $2n - 1$ gives

$$\int_0^1 \pi_n(y) q(y) d\mu(y) = 2 \sum_{j=1}^n w_j \pi_n(y_j) q(y_j) = 0.$$

Thus π_n is the unique monic degree- n orthogonal polynomial for μ , so the nodes are uniquely determined. Exactness for degrees $0, \dots, n - 1$ then determines the weights uniquely through a nonsingular Vandermonde system. Under $y = x^2$, the resulting symmetric $2n$ -node rule is exact for all polynomials of degree at most $4n - 1$, and therefore is the $2n$ -point Gauss–Legendre rule. $\square$

Remark 4. The sets $\Theta_{n,k}$ are nested as moment constraints are added. The selected minimax rules form a discrete hierarchy indexed by k ; no connectedness of the level sets, and no continuous path of minimizers in k , is asserted.

4. Minimax characterization

For $(y, w) \in \mathcal{P}_n$, write

$$q_y(s) = \prod_{j=1}^n (s + y_j), \quad r_{y,w}(s) = \frac{p_{y,w}(s)}{q_y(s)},$$

where $\deg p_{y,w} \leq n - 1$.

Proposition 5 (Difference-numerator degree). *Let $Q_1, Q_2 \in \mathcal{A}_{n,k}$ with $0 \leq k \leq 2n$. Then either their responses are identical or*

$$r_{Q_1}(s) - r_{Q_2}(s) = \frac{H(s)}{q_1(s)q_2(s)}, \quad \deg H \leq 2n - k - 1.$$

For $k = 2n$, the second alternative is impossible.

Proof. The unreduced numerator

$$H = p_1 q_2 - p_2 q_1$$

has degree at most $2n - 1$. For $k = 0$, both responses are $O(s^{-1})$, so their difference is $O(s^{-1})$. For $k \geq 1$, the moment lemma gives

$$r_{Q_1}(s) - r_{Q_2}(s) = O(s^{-k-1}).$$

Since $q_1 q_2$ is monic of degree $2n$, a nonzero polynomial numerator must satisfy

$$\deg H - 2n \leq -k - 1,$$

which is the claimed bound. $\square$

Define

$$\varepsilon_Q(s) = \frac{r_Q(s)}{F(s)} - 1, \quad \|\varepsilon_Q\|_{\infty, S} = \max_{s \in S} |\varepsilon_Q(s)|.$$

Theorem 6 (Class-specific alternation certificate). *Let $0 \leq k < 2n$ and let $Q^* \in \mathcal{A}_{n,k}$. Suppose there are*

$$m = 2n - k + 1$$

distinct ordered points

$$s_1 < \cdots < s_m \quad \text{in } S$$

and a number $E > 0$ such that

$$\varepsilon_{Q^*}(s_i) = (-1)^i \sigma E, \quad \sigma \in \{-1, 1\},$$

and

$$E = \|\varepsilon_{Q^*}\|_{\infty, S}.$$

Then no rule in $\mathcal{A}_{n,k}$ has strictly smaller uniform relative error on S .

Proof. Assume that $Q \in \mathcal{A}_{n,k}$ satisfies

$$\|\varepsilon_Q\|_{\infty, S} < E.$$

At every s_i , the difference

$$\varepsilon_{Q^*}(s_i) - \varepsilon_Q(s_i)$$

has the sign of $\varepsilon_{Q^*}(s_i)$. These signs alternate, so continuity gives at least $m - 1 = 2n - k$ distinct zeros in the open subintervals (s_i, s_{i+1}) . Since $F(s) > 0$ on S , these are zeros of $r_{Q^*} - r_Q$. The difference-numerator proposition allows at most $2n - k - 1$ distinct zeros unless the responses are identical. In the identical-response case, uniqueness of the ordered positive representation gives $Q = Q^*$, which is incompatible with strict improvement. This contradiction proves the result. $\square$

Remark 7. The theorem is a sufficient global-minimax certificate within $\mathcal{A}_{n,k}$. It does not prove necessity of alternation, uniqueness among equal-norm minimizers, or existence of an exact equioscillating parameter vector from floating-point residuals. The endpoint $k = 2n$ is zero-dimensional and is fixed by its moments.

5. Numerical realization of the model-aware hierarchy

The theorem above is conditional on exact moment constraints and exact equioscillation. Numerically, we solve the corresponding nonlinear systems at high precision and report residual audits. We denote the ideal mathematical rule by $Q_{n,k}^*$ and its published decimal operator by $\tilde{Q}_{n,k}$. The table and figures use the computed approximations; they are numerical evidence for the conditional theorem, not a computer-assisted existence proof.

The nonlinear systems impose the moment equations, alternating equal-magnitude errors, and stationarity at internal extrema. Lower-half rules were obtained first, and the upper-half problems were continued toward the Gaussian endpoint. All reported operators have ordered nodes and positive weights.

Table 1: Complete $n = 4$ hierarchy. The endpoint $k = 8$ is zero-dimensional; therefore the alternation column is not applicable.

k	Dimension	Exactness	Alt. points	Worst relative error	Min. w_j
0	8	—	9	1.490623e-10	0.039147
1	7	1	8	2.460626e-09	0.044528
2	6	3	7	4.048641e-08	0.051670
3	5	5	6	6.653546e-07	0.061646
4	4	7	5	1.092819e-05	0.076595
5	3	9	4	1.794333e-04	0.101407
6	2	11	3	2.943219e-03	0.149011
7	1	13	2	4.597982e-02	0.126194
8	0	15	—	4.300499e-01	0.101229

For $n = 4$, the error grows from 1.49×10^{-10} at $k = 0$ to 4.30×10^{-1} at the Gauss endpoint. The upper-half values $k = 5, 6, 7$ continue the approximately constant finite-range price per additional moment before the error enters an order-one regime. This pattern is empirical and is not claimed as an asymptotic theorem.

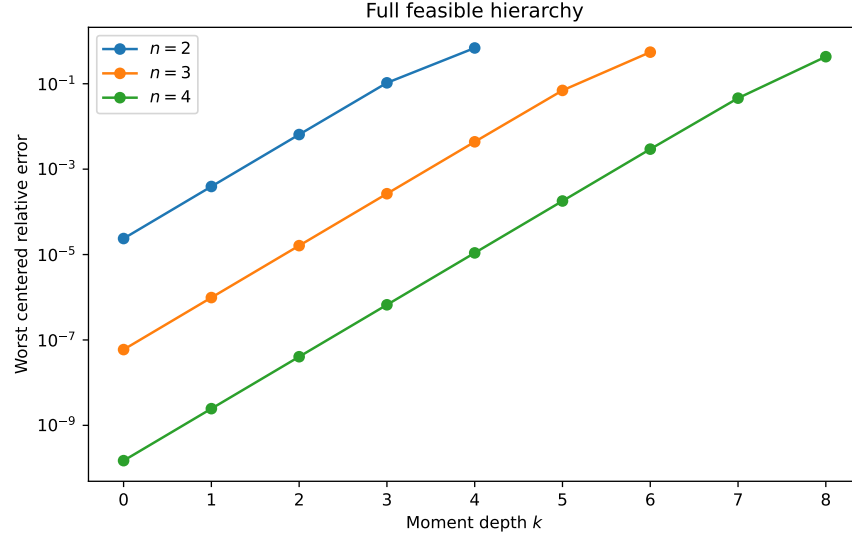


Figure 1: Worst centered relative error for the full feasible hierarchies $0 \leq k \leq 2n$, $n = 2, 3, 4$. The final point of each curve is the corresponding Gauss-Legendre rule.

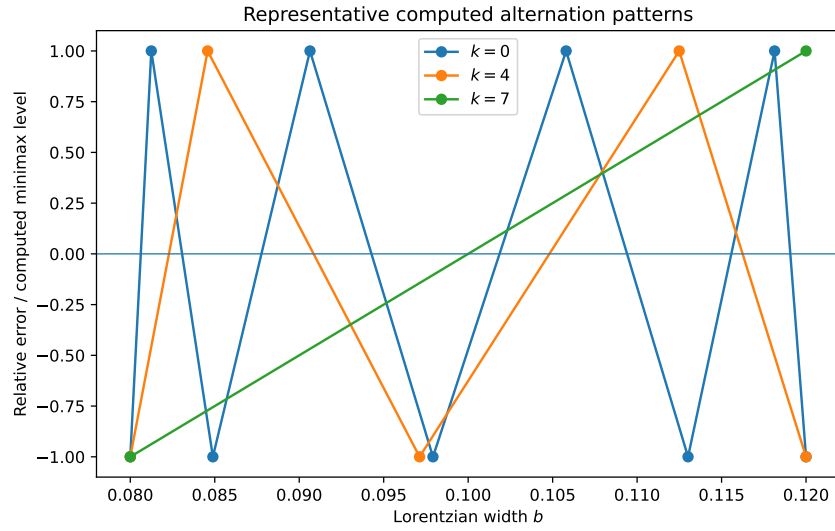


Figure 2: Representative normalized computed alternation patterns from the lower endpoint ($k = 0$), midpoint ($k = 4$), and upper half ($k = 7$) of the $n = 4$ hierarchy.

6. Exact coefficient-space rule selection

For $C \geq 0$ and $\rho > 1$, define

$$\mathcal{H}(C, \rho) = \left\{ h : h(x) = \sum_{j=0}^{\infty} a_j T_j(x), \quad |a_j| \leq C \rho^{-j} \right\}.$$

Because $|T_j(x)| \leq 1$ on $[-1, 1]$ and $\sum_j \rho^{-j} < \infty$, every series in this definition converges absolutely and uniformly and therefore represents a continuous function.

For a fixed bounded quadrature functional Q , define

$$\Delta_j(Q) = I[T_j] - Q[T_j]$$

and

$$G_Q(\rho) = \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

The series converges because

$$|\Delta_j(Q)| \leq \|I - Q\|_{C[-1,1] \rightarrow \mathbb{R}}.$$

Theorem 8 (Exact coefficient-box norm). *For every fixed bounded quadrature functional Q ,*

$$\sup_{h \in \mathcal{H}(C, \rho)} |I[h] - Q[h]| = CG_Q(\rho).$$

Proof. Uniform convergence permits termwise application of the bounded functional $I - Q$, so

$$(I - Q)h = \sum_{j=0}^{\infty} a_j \Delta_j(Q).$$

The coefficient bounds give

$$|(I - Q)h| \leq C \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

Equality is attained by choosing

$$a_j = C \rho^{-j} \operatorname{sign} \Delta_j(Q),$$

with any value in $[-C \rho^{-j}, C \rho^{-j}]$ when $\Delta_j(Q) = 0$. The resulting series belongs to $\mathcal{H}(C, \rho)$. $\square$

For $c_{\max} \geq 0$, define the parameterized mixed class

$$\mathcal{F}(C, \rho, c_{\max}) = \{h + cL_b : h \in \mathcal{H}(C, \rho), |c| \leq c_{\max}, b \in B\}.$$

For a fixed rule, let

$$S_Q = \max_{b \in B} |I[L_b] - Q[L_b]|.$$

The maximum exists because B is compact and the error is continuous in b .

Theorem 9 (Exact mixed-class risk). *For every fixed bounded quadrature functional Q ,*

$$\sup_{f \in \mathcal{F}(C, \rho, c_{\max})} |I[f] - Q[f]| = CG_Q(\rho) + c_{\max} S_Q.$$

Proof. The upper bound follows from the triangle inequality and the preceding theorem. Choose $b_\star \in B$ attaining S_Q . Select $h_\star \in \mathcal{H}(C, \rho)$ such that

$$(I - Q)h_\star = CG_Q(\rho),$$

and choose

$$c_\star = c_{\max} \operatorname{sign}(I[L_{b_\star}] - Q[L_{b_\star}]).$$

Then both error contributions have the same sign and their absolute values add, proving attainability. $\square$

Let $Q_0, \dots, Q_{2n}$ be the finite hierarchy of candidate rules. When $C > 0$, put $\tau = c_{\max}/C$. The exact worst-case selector within this candidate set is

$$k^\star(\rho, \tau) \in \arg \min_{0 \leq k \leq 2n} \{G_{Q_k}(\rho) + \tau S_{Q_k}\}.$$

Remark 10. This selector is exact for the stated coefficient box and structural family, but only among the precomputed candidates $\{Q_0, \dots, Q_{2n}\}$. It does not assert global optimality over all quadrature rules. The rules were constructed by a relative structural minimax objective, whereas S_Q is the absolute structural error appropriate to the additive mixed class.

Table 2: Full $n = 4$ structural and smooth-risk trade-off at $\rho = 1.5$.

k	Structural screening margin S_k	$G_k(1.5)$
0	5.55627e-09	9.11242e-01
1	9.17189e-08	3.95465e-01
2	1.50911e-06	2.04657e-01
3	2.48007e-05	9.59856e-02
4	4.07341e-04	4.49242e-02
5	6.68824e-03	2.14018e-02
6	1.09706e-01	1.17665e-02
7	1.71386e+00	6.62192e-03
8	1.60298e+01	3.42291e-03

7. Conservative numerical screening for the rounded rules

The exact risk theorems apply to each fixed operator. Published decimal nodes and weights, however, have rounding defects, and the numerical screening described here is not an interval or proof-assistant certification.

We re-evaluated all nine rounded $n = 4$ operators. Computed alternation counts were recovered for $k = 0, \dots, 7$. Structural screening margins were formed from a dense midpoint partition together with locally sampled derivative estimates enlarged by a fixed safety factor. Chebyshev defects were evaluated through degree 500, and the remaining series tail was bounded geometrically. A parameter cell was assigned a winner only when its computed upper screening margin was below the computed lower margins of every competitor.

Under this reproducible screening procedure, a unique winner was resolved on

$$97.9067\%$$

of the rectangular coordinate area in

$$(\rho, \log_{10} \tau) \in [1.05, 5] \times [-12, 12].$$

The remaining 2.0933% is concentrated near transition curves. This percentage is an area fraction in the chosen $(\rho, \log_{10} \tau)$ coordinates, not a probability statement.

This screening is reproducible but not mathematically rigorous. The reported area depends on the grid, derivative-sampling rule, safety factor, and refinement depth.

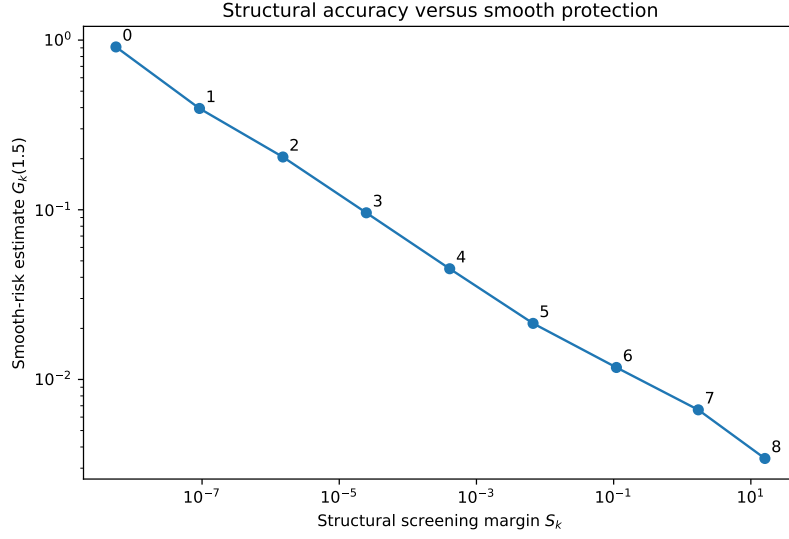


Figure 3: Each added moment decreases the smooth-risk norm but increases the Lorentzian structural error. Labels indicate the moment depth k .

Table 3: Fraction of the resolved screening area assigned to each moment depth.

k	Resolved-area fraction
0	22.279%
1	9.216%
2	8.299%
3	8.463%
4	8.478%
5	8.460%
6	8.117%
7	6.934%
8	19.753%

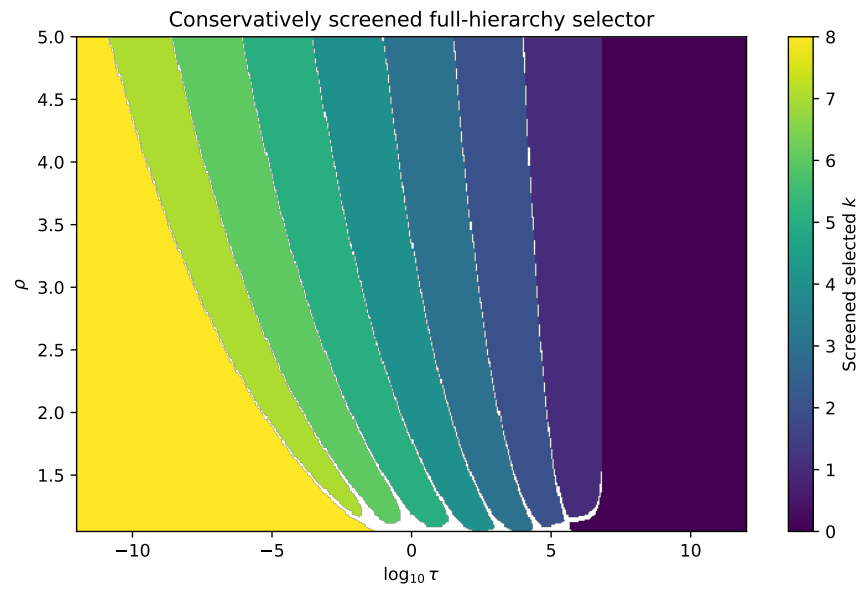


Figure 4: Screened full-hierarchy selector. Blank regions are not assigned because the screening margins overlap.

8. External validity and reconstruction

8.1. Small pole-location mismatch

Without retraining, the centered rules were tested on

$$L_{a,b}(x) = \frac{1}{(x-a)^2 + b^2}, \quad |a| \leq 0.03, \quad 0.08 \leq b \leq 0.12.$$

Table 4: Worst relative errors on the shifted-Lorentzian stress-test box.

Method	Evaluations	Worst error	Worst $ a $	Worst b
PMC-k0	8	4.522336e-06	0.0300	0.0800
PMC-k4	8	1.025564e-03	0.0300	0.0845
Composite-Simpson-9	9	7.426762e-02	0.0300	0.1200
Gauss-Legendre-8	8	4.300499e-01	0.0000	0.0800

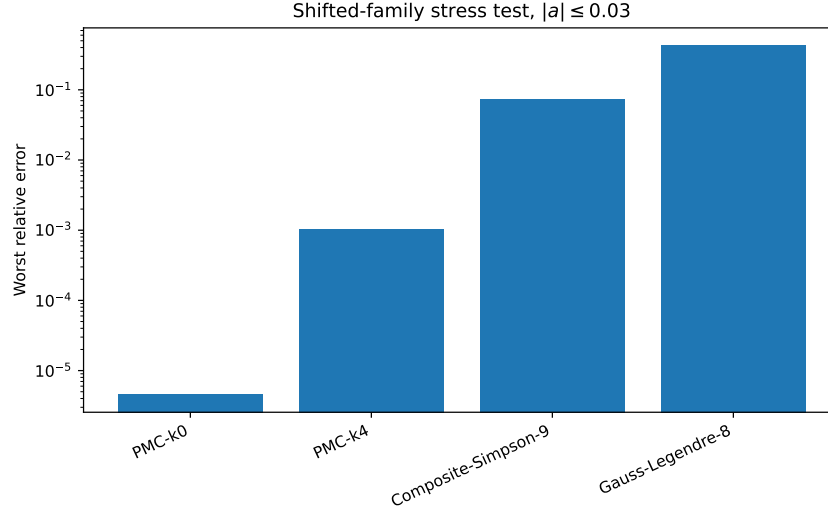


Figure 5: Stress test under small pole-location mismatch. The comparison supports local structural robustness, not shifted-family minimaxity or universal superiority.

The $k = 0$ rule has worst relative error 4.52×10^{-6} , whereas the Gauss endpoint has worst error 4.30×10^{-1} on the same box. The centered error is nevertheless inflated by more than four orders of magnitude, so the correct interpretation is local robustness rather than shift invariance.

8.2. Multi-start reconstruction

Using the same equioscillation equations and solver implementation, the six upper-half problems

$$(2, 3), (3, 4), (3, 5), (4, 5), (4, 6), (4, 7)$$

were reconstructed from twelve perturbed starts each. A run was accepted only when nodes were ordered, weights were positive, moment residuals were below 10^{-10} , and scaled equioscillation residuals were below 10^{-8} .

Table 5: Multi-start reconstruction of the upper-half rules. Error differences are relative to the published minimax level.

(n, k)	Accepted starts	Error difference	Max. node difference	Max. weight difference
(2, 3)	12/12	2.64e-16	5.55e-17	5.55e-17
(3, 4)	10/12	6.78e-14	4.44e-16	5.00e-16
(3, 5)	12/12	5.31e-14	1.29e-14	1.32e-14
(4, 5)	8/12	2.38e-12	4.44e-15	4.27e-15
(4, 6)	11/12	9.22e-14	1.03e-14	1.05e-14
(4, 7)	12/12	5.11e-13	8.95e-14	6.43e-14

All accepted solutions in each problem formed a single error-level cluster. No accepted start found a better rule by more than 10^{-6} relative, and the maximum best-rule error difference was 2.38×10^{-12} . This is a sensitivity test with perturbed initializations. It reduces the risk of an isolated local numerical artifact, but it is neither an independent implementation nor a proof of global uniqueness.

9. Relation to prior work and novelty scope

9.1. Structure-aware and generalized quadrature

Gautschi’s rational Gauss-type rules establish the principle that a quadrature formula can integrate rational functions selected to match important external poles [1]. The present work therefore does not introduce pole-aware quadrature. Its structural objective is different: it minimizes relative error over a continuum of Lorentzian widths instead of enforcing exactness at a finite set of prescribed poles.

Generalized Gaussian quadrature for Chebyshev systems supplies positive interior-node rules exact for non-polynomial spaces, and complete Chebyshev systems support stepwise increases in exactness [2]. Positive exact formulas

and positive-node geometry are also well developed [6, 7]. Accordingly, neither positivity nor an ordered hierarchy of exactness is claimed as new. Here the moment depth indexes a family selected by a continuum Stieltjes minimax criterion at fixed node-pair count.

Optimization-based nested quadrature shows that nonlinear moment-matching systems can be solved flexibly [8]. Our nonlinear solver and multi-start study are reproducibility mechanisms, not the principal conceptual contribution.

9.2. Rational approximation and minimax constraints

Rational minimax approximation and equioscillation are classical; adaptive barycentric algorithms make high-order problems computationally accessible [3]. Interpolation-constrained rational minimax approximation already includes prescribed function values and higher-order derivative conditions [4]. After the change of variable $z = 1/s$, matching Laurent coefficients at infinity is closely related to confluent interpolation at $z = 0$. Thus the novelty claim cannot be “minimax plus constraints.” The alternation theorem in this paper is instead a concise specialization to positive Stieltjes-response quadrature, where the shared coefficients reduce the difference-numerator degree.

Markov-function rational approximation provides relevant relative-error context [9]. Horning and Trefethen make the link from approximation to quadrature explicit by deriving quadrature from rational approximations of Cauchy transforms; their framework also connects Gauss quadrature with Padé approximation at infinity [5]. This is the closest conceptual comparator. The distinction here is the simultaneous enforcement of positive residues, a variable polynomial-moment prefix, a continuum relative-minimax objective, and an exact mixed-class selector.

9.3. Worst-case rule selection

Worst-case analysis for analytic function classes is an established part of quadrature theory [10, 11], and recent positive-weight kernel quadrature gives a different convex-geometric worst-case framework [12]. The coefficient-box identity used here is elementary duality. Its role is to make the smooth residual commensurate with the Stieltjes structural error, producing

$$G_k(\rho) + \tau S_k.$$

The paper’s practically distinctive output is therefore not a new norm identity in isolation, but a selector and screened decision map over a finite family of positive structural rules.

9.4. Defensible novelty statement

The targeted audit found no direct source combining all of the following: positive symmetric Stieltjes residues, continuum Lorentzian relative minimax, variable moment depth at fixed node-pair count, an exact mixed smooth/structural selector, and a validated parameter partition. The defensible claim is this combined model-aware framework. The paper does not claim to invent positivity, moment matching, structure-aware quadrature, constrained rational minimax, equioscillation, or the Gauss/Padé connection separately.

10. Limitations

The theoretical and numerical study is one-dimensional and symmetric. The principal structural family is the centered Lorentzian kernel. The shifted-family experiment is a stress test and is not itself minimax-optimized.

The alternation theorem is conditional: floating-point residuals do not constitute a computer-assisted proof that an exact equioscillating rule exists. Global uniqueness among equal-norm minimizers is not proved. The multi-start study uses the same nonlinear equations and solver implementation, so it is not independent replication.

The selector map is a conservative numerical screening, not rigorous interval validation. Its structural margins use dense sampling and inflated local derivative estimates rather than verified interval bounds. The resolved area, 97.9067%, is measured in the selected $(\rho, \log_{10} \tau)$ coordinates and leaves approximately 2.0933% unassigned. A rigorous computer-assisted selector certificate would require interval bounds for the structural maxima, Chebyshev defects, and all transition cells.

The empirical near-geometric price per moment is not an asymptotic theorem. The literature audit is targeted rather than exhaustive, and equivalent formulations may exist under moment-space, generalized Gaussian, constrained Markov, or interpolatory rational terminology. The manuscript therefore avoids “first” and “unprecedented” claims.

11. Conclusion

Within the stated positive symmetric Stieltjes class, the variable moment depth organizes a path

$$k = 0 \quad \longrightarrow \quad k = 2n$$

from a structural minimax rule to Gauss–Legendre. The feasible-range characterization and difference-degree argument support the numerical construction, while the exact mixed risk turns the family into a model-aware selector. Complete computations for $n = 2, 3, 4$, conservative decision-map screening, a shifted-pole stress test, and multi-start reconstruction quantify both the promise and the limits of the approach.

The main contribution is not any ingredient separately. It is the combination of positive Stieltjes realization, continuum structural minimax, variable polynomial protection, and exact rule selection in one computable framework. The next scientific step is independent specialist comparison with the closest generalized-quadrature and constrained-rational literature, followed by extensions beyond symmetry and the centered kernel.

Data and code availability

The code, published rules, generated data, and reproduction scripts are available in the public GitHub repository. The Version 2.2 theorem-audit and full-selector materials are included in the accompanying research archive and should be merged into the public repository before journal submission.

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During the preparation of this work, the author used OpenAI ChatGPT to assist with literature organization, code prototyping, manuscript structuring, and language revision. The author reviewed and verified the mathematical derivations, numerical results, references, interpretations, and final wording and takes full responsibility for the content of the article.

References

- [1] W. Gautschi, Gauss-type quadrature rules for rational functions, arXiv:math/9307223 (1993).
- [2] D. Huybrechs, On the computation of Gaussian quadrature rules for Chebyshev sets of linearly independent functions, arXiv:1710.11244 (2017).
- [3] S.-I. Filip, Y. Nakatsukasa, L. N. Trefethen, B. Beckermann, Rational minimax approximation via adaptive barycentric representations, arXiv:1705.10132 (2017).
- [4] L.-H. Zhang, Y.-N. Zhang, b-d-Lawson: a method for the interpolation constrained rational minimax approximation, arXiv:2502.10665 (2025).
- [5] A. Horning, L. N. Trefethen, Quadrature formulas from rational approximations, arXiv:2507.14971 (2025).
- [6] L. M. M. van den Bos, B. Sanderse, A geometric approach for the addition of nodes to an interpolatory quadrature rule with positive weights, arXiv:1902.07477 (2019).
- [7] J. Glaubitz, Constructing positive interpolatory cubature formulas, arXiv:2009.11981 (2020).
- [8] V. Keshavarzzadeh, R. M. Kirby, A. Narayan, Generation of nested quadrature rules for generic weight functions via numerical optimization: Application to sparse grids, arXiv:1808.03707 (2018).
- [9] B. Beckermann, J. Bisch, R. Luce, On the rational approximation of Markov functions, with applications to the computation of Markov functions of Toeplitz matrices, arXiv:2106.05098 (2021).
- [10] L. N. Trefethen, Exactness of quadrature formulas, arXiv:2101.09501 (2021).
- [11] T. Goda, Y. Kazashi, K. Tanaka, How sharp are error bounds? Lower bounds on quadrature worst-case errors for analytic functions, arXiv:2401.07196 (2024).

- [12] S. Hayakawa, Convex-geometric error bounds for positive-weight kernel quadrature, arXiv:2605.05705 (2026).
- [13] R. Nailwal, A. Zalar, Gaussian quadratures with prescribed nodes via moment theory, arXiv:2412.20849 (2024).